

1. Problem Tackled (Recognizer, Decider, or Computation?)

This machine is a **decider** for the language:

$$L = \{ w \in \{a,b\}^* \mid w \text{ begins with the letter 'a'} \}$$

The machine always halts with either **qaccept** or **qreject**, so it is a decider.

2. Reference for the Problem

There wasn't any specific reference for this problem, but I took inspiration from questions I did in the past from Theory of Computation, 3rd edition by Michael Sipser

3. How the Machine Works (in words)

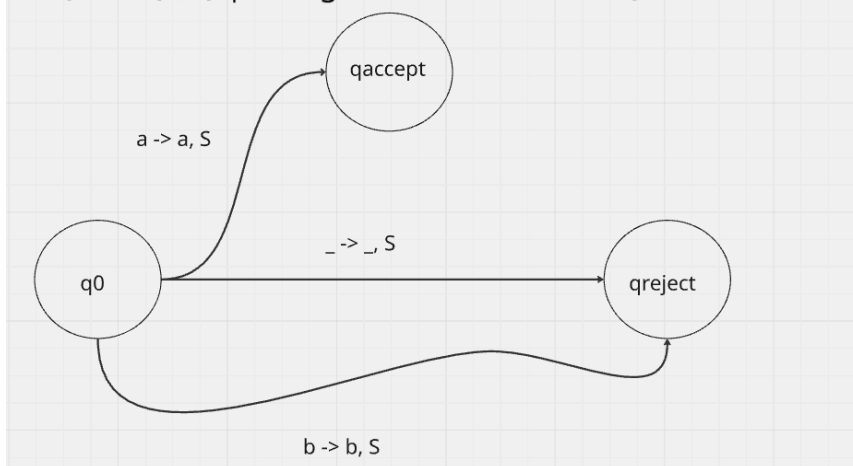
The machine immediately checks the **first symbol** on the tape.

- If the first symbol is '**a**', it **accepts**.
- If the first symbol is '**b**' or blank '**_**' (i.e., empty string), it **rejects**.

It does not move its head and does not need to inspect the rest of the tape. The entire decision is made from the first tape cell. The machine halts in one step for all inputs.

4. State Diagram

1Tape Turing Machine Decider that accepts if first letter is a
 $L = \{ w \in \{a,b\}^* \mid w \text{ begins with the letter 'a'} \}$



5. Verifying correct operation

I verified the machine by running several test inputs and checking that the results matched the rule “accept if the first symbol is **a**, reject if it is **b**.” The inputs **aabb** and **aba** both begin with a , and the machine accepted them as expected. The inputs **ba** and **baa** begin with b , and the machine correctly rejected both. I also tested **abc** and **cba** to confirm behavior with other symbols: **abc** was accepted because it starts with a , while **cba** was rejected because it starts with c and there is no accepting rule for that case.